

DATA ENGINEERING AND EXPLORATORY DATA ANALYSIS PIPELINE

Project 1 – Implementation Report

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Project: E-Commerce Transaction Data Preprocessing

Architecture: Input–Process–Output (IPO)

Framework: Python, Pandas, NumPy, and Scikit-learn

1. Executive Summary

This project focuses on developing a structured data preprocessing pipeline for an e-commerce transaction dataset. The primary objective is to transform the raw dataset into a clean, consistent, and machine-learning-ready format.

The preprocessing workflow addresses several common data quality issues, including missing values, numerical outliers, feature engineering, categorical variables, and multicollinearity. Each step was performed systematically to maintain the quality and consistency of the original records while preparing the data for further analysis and predictive modeling.

The final processed dataset was validated to ensure that missing values and unnecessary redundant features were properly handled. The cleaned dataset was then exported as “Cleaned_Dataset_Project1.csv” for use in subsequent machine learning and analytical tasks.

2. Project Objectives

The main objectives of this project were to:

- Examine the structure and quality of the raw e-commerce dataset.
- Identify and appropriately handle missing values.
- Detects and manages extreme numerical observations.
- Create meaningful features from existing variables.
- Convert categorical variables into a machine-readable format.
- Identify and reduce highly correlated predictor variables.
- Validate the processed dataset.
- Export the final cleaned dataset for downstream analysis and machine learning.

3. Dataset Overview

The dataset contains e-commerce transaction records with information related to customer orders, products, payments, discounts, and transaction values.

The major variables considered during preprocessing include:

Feature	Description
Date	Date on which the transaction occurred
Product	Product associated with the transaction
Quantity	Number of items purchased
UnitPrice	Price of one item
TotalPrice	Total transaction value
ItemsInCart	Number of items in the customer's cart
CouponCode	Coupon used during the transaction
PaymentMethod	Method used for payment
OrderStatus	Current status of the order
ReferralSource	Source through which the customer reached the store

The dataset was first examined to understand its structure, data types, missing values, and potential anomalies before applying any transformations.

4. Methodology

The complete preprocessing workflow follows an **Input–Process–Output (IPO)** architecture.

Input

Raw e-commerce transaction dataset.

Process

1. Data inspection
2. Missing value treatment
3. Outlier detection and treatment
4. Feature engineering
5. Categorical encoding
6. Correlation analysis
7. Removal of redundant features
8. Final validation

Output

A clean and structured dataset suitable for exploratory analysis and machine learning.

5. Data Inspection and Quality Assessment

Phase 1: The raw dataset was initially loaded and examined to understand its overall structure and identify potential data quality problems.

The following aspects were checked:

- Number of rows and columns
- Data types
- Missing values
- Numerical distributions
- Categorical variables
- Potential outliers
- Relationships between numerical features

This initial inspection provided the basis for selecting appropriate preprocessing techniques for each variable.

5.1 Missing Data Handling

Missing values can affect statistical analysis and may cause errors in machine learning algorithms. Therefore, missing values were examined across the dataset before further processing.

The **CouponCode** feature contained a relatively high proportion of missing values, exceeding the 20% threshold considered for this project.

Instead of removing the corresponding records, the missing entries were replaced with: **NoCoupon**

This approach preserves the original transaction records while clearly indicating that no coupon code was provided.

Reason for this approach

Removing rows with missing coupon information could unnecessarily reduce the size of the dataset. Since the absence of a coupon is meaningful information in this case, replacing the missing values with **NoCoupon** provides a clear and interpretable representation.

5.2 Outlier Detection and Treatment

Outliers are observations that differ substantially from the general pattern of a dataset. Extreme values can affect statistical measures and may negatively influence certain machine learning models.

For numerical variables, the **Interquartile Range (IQR)** method was used to identify potential outliers.

The IQR is calculated as:

$$\text{IQR} = Q3 - Q1$$

The lower and upper boundaries were determined using:

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

Values outside these boundaries were considered potential outliers.

Outlier Detection Result

The analysis identified eight extreme observations in the **TotalPrice** feature.

Rather than deleting the complete records, the extreme values were capped within the calculated IQR boundaries using `numpy.clip()`.

Reason for Capping

Removing complete transaction records could result in the loss of useful information from other features. Capping limits the influence of extreme values while retaining the corresponding records in the dataset.

This approach therefore provides a balance between reducing the effect of extreme observations and preserving the overall dataset size.

6. Feature Engineering

Phase 2: Feature engineering was performed to create additional variables that could provide useful information for subsequent analysis and predictive modeling.

Three new features were created.

6.1 UsedCoupon

The **UsedCoupon** feature indicates whether a coupon was used for a transaction.

- 1 → Coupon was used
- 0 → No coupon was used

This converts the information contained in CouponCode into a simple numerical indicator that can be directly used by machine learning algorithms.

6.2 OrderMonth

The **OrderMonth** feature was extracted from the Date variable.

The date was first converted into a proper datetime format using Pandas, after which the month was extracted.

For example: **Date** → **Month**

2025-05-18 → 5

This feature can help identify possible monthly or seasonal patterns in transaction activity.

6.3 AvgItemValue

The **AvgItemValue** feature was calculated using the following formula:

AvgItemValue = TotalPrice / Quantity

This feature represents the average monetary value associated with each item in a transaction.

It provides an additional numerical representation of transaction value that can be useful during exploratory analysis.

6.4. Categorical Data Encoding

Several variables in the dataset are categorical rather than numerical. These include:

- Product
- PaymentMethod
- OrderStatus
- ReferralSource

Machine learning algorithms generally require numerical input. Therefore, these categorical variables were converted into numerical representations using **One-Hot Encoding**.

The encoding was performed using Pandas' `get_dummies()` function.

For example, if PaymentMethod contains:

- Cash
- Card
- Mobile Banking

The variable can be represented using separate binary columns.

This prevents the model from incorrectly interpreting categories as having a numerical order. For example, assigning Cash = 1, Card = 2, and Mobile Banking = 3 would incorrectly imply that Mobile Banking is greater than Card.

One-Hot Encoding avoids this problem by representing each category independently.

7. Multicollinearity Analysis

Phase 3: After feature engineering and categorical encoding, the numerical predictor variables were examined for strong correlations.

The numerical features considered included:

- Quantity
- UnitPrice
- ItemsInCart
- AvgItemValue

A correlation matrix was generated to identify highly correlated feature pairs.

For this project, an absolute correlation coefficient greater than **0.80** was considered an indication of strong correlation.

Why Multicollinearity Matters

Multicollinearity occurs when two or more predictor variables contain highly similar information.

High multicollinearity can make regression coefficients difficult to interpret and may reduce the stability of statistical models.

Therefore, highly redundant features were reviewed before preparing the final dataset.

7.1. Identification of Redundant Feature

The correlation analysis showed a **perfect correlation ($r = 1.0$)** between:

- AvgItemValue
- UnitPrice

This relationship is expected from the structure of the dataset because:

$$\text{AvgItemValue} = \text{TotalPrice} / \text{Quantity}$$

and, for the transactions represented in the dataset, this calculation produces the same information as UnitPrice.

Therefore, keeping both variables would provide duplicate information to the model.

Final Decision

AvgItemValue was removed from the final feature set, while UnitPrice was retained.

This reduced redundancy without removing useful information from the dataset.

8. Data Validation

After completing all preprocessing operations, the resulting dataset was validated before export.

The validation process checked that:

- Missing values had been appropriately handled.
- Numerical variables were within the expected ranges after outlier treatment.
- Categorical variables had been properly encoded.
- Redundant features had been removed.
- The final dataset had a consistent structure.
- The processed features were suitable for downstream analysis.

Runtime assertions were also used to verify important conditions, including the absence of remaining missing values in the final feature set.

9. Final Output

After completing the preprocessing pipeline, the cleaned dataset was exported as:

Cleaned_Dataset_Project1.csv

The final dataset provides a structured representation of the original e-commerce transactions and is suitable for:

- Exploratory Data Analysis (EDA)
- Statistical analysis
- Data visualization
- Machine learning model development
- Predictive analysis

10. Pipeline Summary

The overall preprocessing workflow can be summarized as:

Raw E-Commerce Dataset

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Data Inspection

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Missing Value Treatment

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Outlier Detection and Capping

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Feature Engineering

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Categorical Encoding

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Correlation Analysis

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Removal of Redundant Features

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Data Validation

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Cleaned Dataset

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Cleaned_Dataset_Project1.csv

11. Conclusion

The project successfully implemented a structured preprocessing pipeline for the e-commerce transaction dataset. The raw data was examined and processed to address missing values, extreme observations, categorical variables, and redundant numerical features.

Missing coupon information was handled using a meaningful category, while extreme TotalPrice values were capped using the IQR method to reduce their influence without removing complete transaction records. Additional features were created to improve the representation of transaction and temporal information. Categorical variables were converted using One-Hot Encoding, and correlation analysis was used to identify and remove redundant predictors.

The final dataset was validated and exported in CSV format. Overall, the preprocessing pipeline provides a reliable foundation for further **exploratory data analysis and machine learning applications**.

12. Tools and Technologies Used

Tool / Library	Purpose
Python	Main programming language
Pandas	Data loading, cleaning, transformation, and analysis
NumPy	Numerical operations and outlier capping
Matplotlib/ Seaborn	Data visualization and exploratory analysis
Scikit-learn	Machine learning-related preprocessing and analysis
CSV	Final dataset storage format